Question 1: Downloading and installing Hadoop framework; startup scripts and configuration files

startup scripts and configuration files

Aim

To install the Hadoop framework, configure it for pseudo-distributed operation, and start the Hadoop daemons using the

startup scripts.

Procedure

1. Install a supported Java Development Kit and verify it using java -version. Set JAVA_HOME to the JDK installation

directory.

2. Download a Hadoop 3.x binary distribution from the Apache Hadoop website and extract it, for example, into

/usr/local/hadoop.

3. Set HADOOP_HOME and add Hadoop's bin and sbin directories to PATH. Set JAVA_HOME in Hadoop's

hadoop-env.sh.

4. Configure core-site.xml, hdfs-site.xml, mapred-site.xml and yarn-site.xml as shown below.

5. Create the NameNode and DataNode storage directories and ensure the current user has permission to access

them.

6. Format the NameNode once using hdfs namenode -format.

7. Start HDFS using start-dfs.sh and YARN using start-yarn.sh. Check running Java processes with jps.

8. Open the NameNode web interface, usually http://localhost:9870, to verify the HDFS service.

Configuration files

Example settings for a single-machine pseudo-distributed lab setup:

~/.bashrc

export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64

export HADOOP_HOME=/usr/local/hadoop

export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin

<!-- $HADOOP_HOME/etc/hadoop/core-site.xml -->

<configuration>

<property>

<name>fs.defaultFS</name>

<value>hdfs://localhost:9000</value>

</property>

</configuration>

<!-- $HADOOP_HOME/etc/hadoop/hdfs-site.xml -->

<configuration>

<property>

<name>dfs.replication</name>

<value>1</value>

</property>

<property>

<name>dfs.namenode.name.dir</name>

<value>file:///usr/local/hadoop/data/namenode</value>

</property>

<property>

<name>dfs.datanode.data.dir</name>

<value>file:///usr/local/hadoop/data/datanode</value>

</property>

</configuration>

<!-- $HADOOP_HOME/etc/hadoop/mapred-site.xml -->

<configuration>

<property>

<name>mapreduce.framework.name</name>

<value>yarn</value>

</property>

</configuration>

<!-- $HADOOP_HOME/etc/hadoop/yarn-site.xml -->

<configuration>

<property>

<name>yarn.nodemanager.aux-services</name>

<value>mapreduce_shuffle</value>

</property>

</configuration>

Startup scripts / commands

source ~/.bashrc

hdfs namenode -format # Run only for first-time initialization

start-dfs.sh

start-yarn.sh

jps

To stop the services:

stop-yarn.sh

stop-dfs.sh

Output of Program

$ jps

1234 NameNode

1350 DataNode

1480 SecondaryNameNode

1590 ResourceManager

1702 NodeManager

1800 Jps

Result

Hadoop is installed and configured in pseudo-distributed mode. HDFS and YARN daemons are started and verified

using jps and the NameNode web interface.